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Deploying Automated Buffer Production for cGMP Use: Points to Consider

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Abstract

Biopharmaceutical manufacturing often takes place in tank farms — facilities in which large-volume vessels are used to support cell culture processes with equally sized, or even larger buffer preparation and storage tanks to support downstream processing. While the large cell culture vessels used to produce products are justifiable, current downstream buffer management approaches relying on high-capacity tanks lead to constraints on facility construction, operations, and plant flexibility.

Evaluating and then implementing buffer storage and handling alternatives can help to alleviate existing costs and operational challenges. Automated just-in-time (JIT) at point-of-use (POU) buffer delivery using solution concentrates has been evaluated in this paper as one such alternative. The success of this approach is built on well-designed, automated hardware, the associated control algorithms, and proper understanding of the basic chemistry involved in buffer formulation.

From our evaluation studies, chemical equilibria, in conjunction with equipment design parameters have been shown to be critical to the robustness and reliability of automated buffer formulation. By demonstrating the proper understanding and control of these critical parameters, the stage is set for a science-based change management road map for adopting automated JIT at POU buffer management. This alternative to existing manual practices enables organizations to increase throughput and plant flexibility, and reduce plant construction and operational costs while continuing to meet current quality, regulatory, and compliance expectations.

1. Introduction

A typical biopharmaceutical manufacturing plant is composed of media/buffer preparation and storage, cell culture expansion and production, downstream purification, and drug substance (DS) fill areas. In large-scale facilities with production bioreactors sized at 15,000 L or larger, the downstream buffer preparation and storage areas take up significant amounts of total plant footprints. They are equipped with multiple large-volume fixed tanks for solids handling, solution mixing, and subsequent static storage to supply a diversity of buffers and volumes needed to manufacture each batch of DS. In addition, complex networks of stainless steel piping are utilized to transfer solutions from the preparation tanks to the storage tanks and finally to the points of use.

This time-honored approach to buffer handling negatively impacts facility design, construction, and routine operations:

(1) The fixed tanks, piping, and transfer panels expand the scope (from equipment to automation), cost, and schedule when building or upgrading a facility.

(2) Required cleaning and maintenance of the fixed tanks and piping increase operating expenses.

(3) Buffer preparation operations, from solid weigh-out to tank cleaning performed at the end of each DS operation, often cause bottlenecks, limiting plant output.

Figure 1 illustrates that seven out of the top ten pieces of equipment with the longest duration of occupancy are the buffer preparation and storage vessels needed for chromatography column

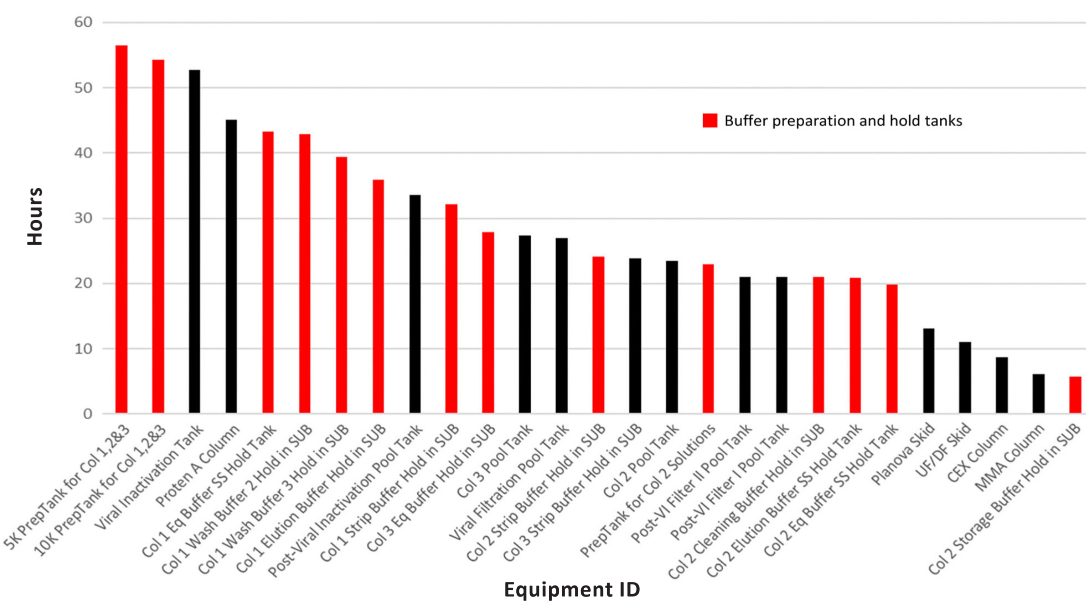

FIGURE 1. Typical equipment utilization duration for a platform mAb batch process.
Legend: Column (Col); equilibration (Eq); cation exchange (CEX); mixed mode anion (MMA) exchange; stainless steel (SS); single-use bioreactor (SUB); viral inactivation (VI); and ultrafiltration/diafiltration (UF/DF).

operations. Protein A chromatography, usually deployed as the initial product capture step in monoclonal antibody processing, tends to be the longest unit operation because it is designed to capture the entire 15,000 L or more of the harvest intermediate into smaller volumes of partially purified material. Vessels used in the preparation and storage of Protein A chromatography buffers are occupied ahead of time, and the storage vessels are tied up until the end of the Protein A chromatography operation. Utilization of these tanks is further prolonged since these resources are shared with other chromatography operations. Therefore, solution tanks in downstream processing tend to cause plant operation bottlenecks.

Recent cell culture productivity improvements that lead to increasing titers present additional demands on buffer preparation operations that cannot be met by simply scaling up standard purification approaches, as higher amounts of cell culture intermediates require higher volumes of buffers. Adding larger preparation and storage tanks to an existing plant is not cost-effective, and in some cases, may not be an option due to space constraints. To meet this challenge, automated JIT at POU buffer management that eliminates the need for a buffer hold entirely is proposed. This approach recognizes that: (1) a buffer is defined by the concentrations of the acid and base conjugate pair that determine the proper buffer capacity at the appropriate pH range; and (2) a buffer can be prepared by a liquid handling system that mixes solutions of the conjugate pair and water-for-injection (WFI) together at the POU.

In-line conditioning (IC)[1] is an automated JIT at POU buffer management technology in which a process buffer is formulated in-line from concentrated stock solutions of acids, bases, salts, and additives that are mixed with the correct amount of WFI. This paper presents an alternative to traditional manual buffer management practices. IC control strategies are designed to prepare buffers automatically via feedback control based on conductivity, pH, flow, or a combination of pH and flow. The identification of effective performance variability sources and the associated chemical and engineering parameters critical to controlling such variabilities are required prior to implementation. The IC process must be deemed equivalent to existing manual practices in order to satisfy quality, regulatory, and compliance standards. Deploying automated buffer production systems to assure precision and control would improve operational costs for the current good manufacturing practice (cGMP) environment.

In this article, the automated IC system used in this study is discussed first and is followed by the buffer preparation performance data collected. Sources of performance variability and critical operating parameters are identified. Then a brief review of chemical equilibria principles is included to further demonstrate our understanding of the factors governing buffer characteristics and system performance in an automated IC environment. Lastly, the benefits of IC technology is discussed, and a preliminary change management road map is proposed.

FIGURE 2. IC system connected to disposable bags with ready-made stock solutions.

2. IC System Description

The IC system (**Figure 2**) used in this study has five 3-headed pumps (Lewa) and flow meters for controlled delivery of acids, bases, WFI, salt solutions, and additives when needed. The system is equipped with multiple pH and conductivity sensors for monitoring and dynamic feedback control. The equipment diagram is shown in **Figure 3**.

Target pH was achieved via two different control scheme experiments: pH-flow and flow feedback. A third operational scheme, pH-conductivity feedback control, can also be utilized with this IC system.[1] However, it was not included in the scope of this particular study.

In the pH-flow feedback mode, one pH signal was used to automatically adjust the flow rates of acid and base to reach the pH setpoint. The rest of the pH signals were used for monitoring only. Flow feedback was still in use to maintain buffer and salt concentrations, and target total volumetric flow rate. In the flow feedback mode, the system used pre-defined recipes to determine the flow setpoint for each pump while ensuring that the calculated total volumetric flow rate was met. Conductivity and pH meters were used for monitoring only. Once the starting

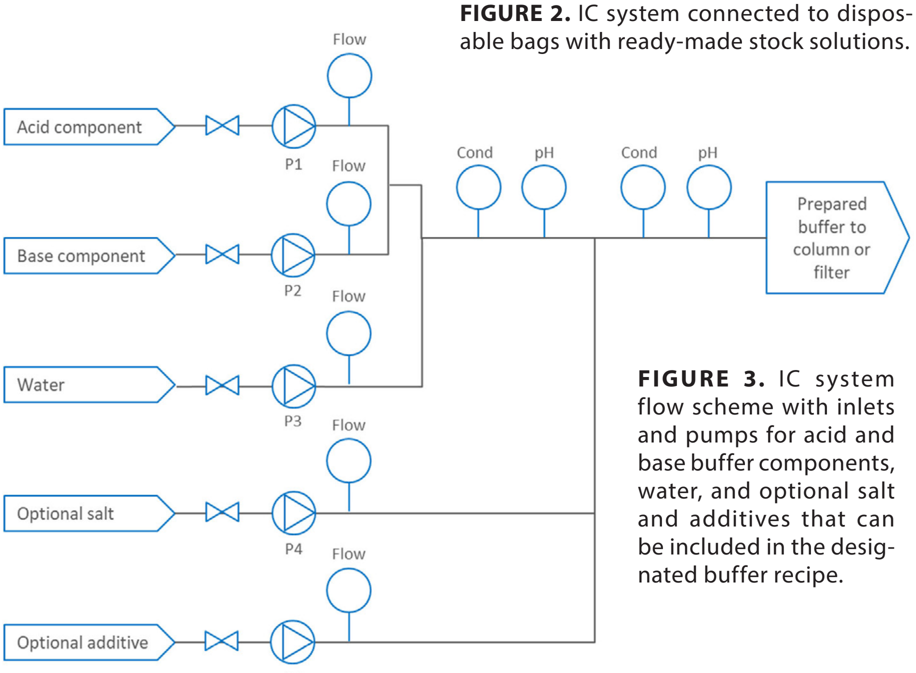

FIGURE 3. IC system flow scheme with inlets and pumps for acid and base buffer components, water, and optional salt and additives that can be included in the designated buffer recipe.

stock solution concentrations were specified, the constant buffer concentration was obtained by appropriate proportioning of the corresponding acid and base flows.

For this study, the operating ranges of the individual pumps were 4–180 L/h for the acid, base, and salt pumps, and 15–600 L/h for WFI. Flow accuracy was maintained at the larger value of either ±1% of the maximum pump flow rate or ±2% of target.

The control strategy employed by the IC skid for real-time buffer formulation is composed of a four-stage process to ensure expected performance:

(1) Short priming of stock solutions at high pump speed

(2) Buffer formulation using flow feedback (~1.0 min)

(3) Activation of pH feedback and search for stability window (~0.5 min)

(4) Buffer release and initiation of buffer verification function (BVF)

BVF monitors the pH and/or conductivity of a mixed buffer. If at least one of the detected values is outside of the acceptance limits, it automatically redirects the buffer out through a main drain or bypasses the column, changing the outlet to waste.

3. Sources of Buffer Variability and Critical Controls in IC

To properly control automated IC capabilities when implemented in a cGMP environment, the process parameters critical in controlling the pH and conductivity of a buffer need to be identified and characterized. Using the design of experiments (DoE) module in UNICORN™ (GE Healthcare) v. 6.3 control software, a set of experiments based on a cubic centered face (CCF) design using an acetate system (with pH 3.7–4.7, buffer concentration 20–100 mM, and salt concentration 0–250 mM, in triplicate at the center point of pH = 4.7, [buffer] = 60 mM and [NaCl] = 125 mM) were conducted to identify the critical process parameters and characterize their impact on buffer pH and conductivity control in an automated IC environment. The experiments were composed of 17 different buffers formulated at 600 L/h under the flow feedback and pH-flow feedback modes with two starting stock solution concentrations: 4 M acetic acid and 4 M sodium acetate, and 1 M acetic acid and 1 M sodium acetate respectively. Using the 4 M starting stock solutions enabled intentional testing of the effect of operating the automated IC system near the lowest flow rate limits on performance. In total, there were 17 × 2 = 34 data points for flow feedback and 34 data points for pH-flow feedback, as depicted in **Table 1**.

3.1 Sources of Buffer pH Variability and Associated Critical IC Control Elements

In **Figure 4**, pH variability data are presented as a function of buffer capacity under flow feedback and pH-flow feedback controls. The data indicate that pH variability is best controlled when preparing a high buffer capacity solution while operating the IC pumps within their design specifications. However, pH variability due to pump variability effect could be mitigated by buffer capacity in some instances (data not shown), indicating that buffer capacity is a dominant critical process parameter. To demonstrate the suitability of this technology, the pH data

TABLE 1. CCF-designed experiments using an acetate system.

Buffer (M)	pH	[NaCl] (M)	Buffer (M)	pH	[NaCl] (M)	Buffer (M)	pH	[NaCl] (M)
0.02	3.7	0	0.06	4.7	0.125	0.1	3.7	0.25
0.02	3.7	0	0.06	4.7	0.125	0.02	5.7	0.25
0.06	4.7	0.125	0.02	3.7	0.25	0.02	5.7	0.25
0.06	4.7	0.125	0.02	3.7	0.25	0.06	4.7	0.25
0.06	3.7	0.125	0.1	5.7	0	0.06	4.7	0.25
0.06	3.7	0.125	0.1	5.7	0	0.1	5.7	0.25
0.06	5.7	0.125	0.06	4.7	0.125	0.1	5.7	0.25
0.06	5.7	0.125	0.06	4.7	0.125	0.1	4.7	0.125
0.02	4.7	0.125	0.02	5.7	0	0.1	4.7	0.125
0.02	4.7	0.125	0.02	5.7	0	0.06	4.7	0
0.1	3.7	0	0.1	3.7	0.25	0.06	4.7	0
0.1	3.7	0						

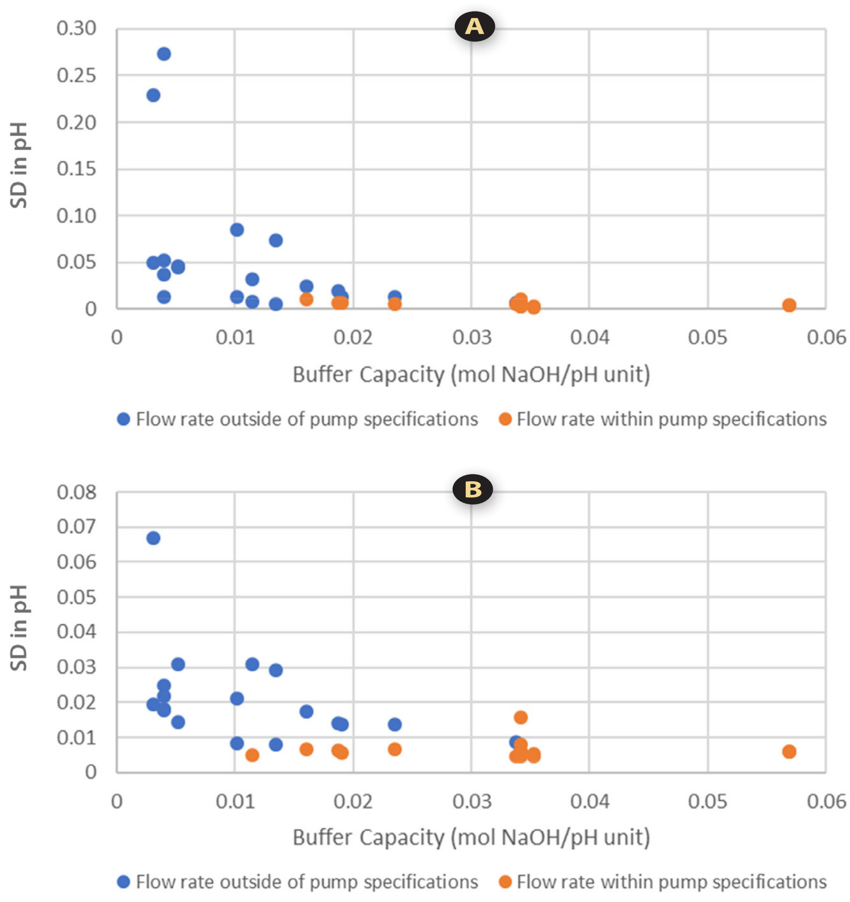

FIGURE 4. pH variability vs. buffer capacity under: (A) pH-flow; and (B) flow feedback controls. (SD = standard deviation)

were analyzed in more detail with respect to a typical buffer release specification of ± 0.1 pH unit. With each of the experiments, differences between target and maximum/minimum pH were observed under different modes of control that were generated with IC pumps operating within their design specifications (**Figures 5** and **6**). In general, buffers prepared under pH-flow feedback control met the ± 0.1 pH unit requirement more consistently than those prepared under flow feedback control.

A detailed review of the experimental data (**Appendix, Table 2**) indicated that, in all but one instance (highlighted in red), IC pumps operated within their specified limits and all buffers met the ± 0.1 pH unit requirement consistently, regardless of the mode of control. It should be noted, however, that buffer pH minimum/maximum distribution tended to be tighter when operating under pH-flow feedback control. However, if the release specification was ± 0.05 pH unit, pH-flow feedback control would be required when using this automated IC technology. In addition, pH-flow feedback control should be capable of mitigating the effects of stock solution concentration variability, which was not explicitly tested in this study. The data collected from these experiments identified buffer capacity and IC pump flow rate as the critical parameters in controlling buffer pH when using automated IC, and process robustness was maximized when making a buffer with high buffer capacity while operating the IC pumps within their design specifications.

This finding is not surprising and confirms the known fact that buffer preparation is dependent on how well a solution is designed around the pK_a of the conjugate pair (*i.e.*, the effect of buffer capacity) and how accurately one can measure the amount of individual components that make up the final buffer. With automated IC, where solutions rather than solids are used for buffer formulation, accurate measurement of individual components is achieved by precise solvent delivery with the IC pumps.

3.2 Sources of Buffer Conductivity Variability and Associated Critical IC Control Elements

In **Figure 7** (on page 5) the SD for buffer conductivity derived from the time-course data is presented as a function of buffer capacity under flow or pH-flow feedback control. Similar to observations from the pH profiles, buffer conductivity variability is also best controlled for a solution with high buffer capacity prepared by operating the pumps within their design specifications. Since buffer conductivity is not dependent on buffer capacity, this apparent dependence on capacity is likely a secondary effect resulting from less instantaneous adjustments of the ionic species concentrations while the IC system is attempting to maintain target pH of a high buffer capacity

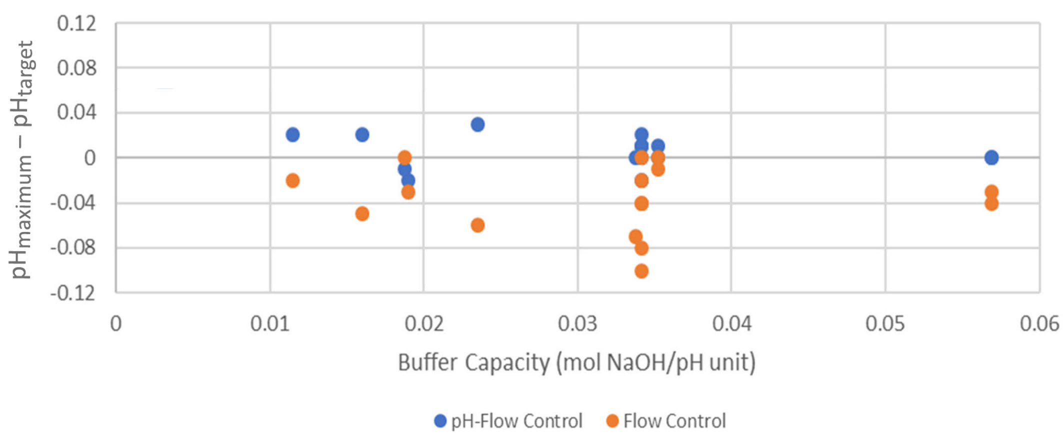

FIGURE 5. ($pH_{maximum} - pH_{target}$) vs. buffer capacity under pH-flow and flow feedback controls.

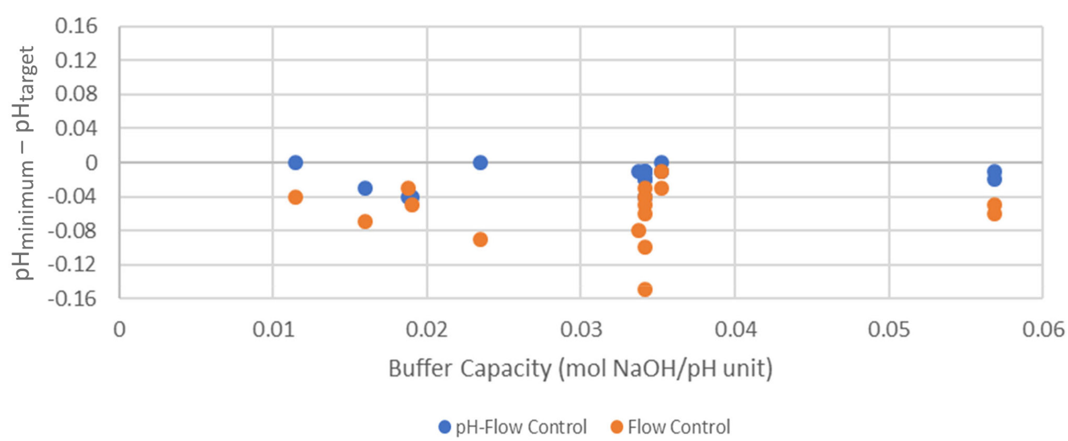

FIGURE 6. ($pH_{minimum} - pH_{target}$) vs. buffer capacity under pH-flow and flow feedback controls.

solution. On the other hand, the magnitude of the conductivity SD becomes more pronounced at decreasing buffer capacity, potentially because of more frequent system adjustments to maintain target pH, or more instantaneous perturbations to the ionic species concentrations of a low buffer capacity solution.

To further demonstrate the fitness of automated IC for buffer preparation, the conductivity data were analyzed in more detail with respect to a typical release specification. Minimum/maximum conductivity values, with respect to the range of ±10% of target under different modes of control generated with pumps operating within their design specifications are provided in **Figure 8** and **Figure 9** (on page 6).

A detailed review of the buffer conductivity data from each experiment is listed in **Table 3** of the **Appendix** section. When operating the IC technology within the pump specification limits, each buffer prepared meets the ±10% of target requirements consistently, regardless of the control mode. The experimental data consistently identify pump flow rate to be the critical parameter in controlling buffer conductivity when using the IC technology. Process robustness was maximized when the IC pumps ran within design specifications while benefiting from lower level perturbation of the ionic species concentration in the process.

4. Chemical Equilibria and Buffer Capacity Calculation

Since buffer capacity has been identified as one of the critical parameters in determining automated IC performance, the ability to accurately predict buffer capacity of an electrolyte solution is expected in a cGMP process. This can be achieved by applying the Kohlrausch equation[2] to predict conductivity after: (1) correcting for the finite concentration effect of an electrolyte solution using the Debye-Hückel theory[3]; and (2) using the Henderson-Hasselbach equation[4] to predict solution pH. Accounting for overall mass balance and charge neutrality, the buffer capacity can be accurately calculated using the system of equations provided in the **Appendix**. The outcome of this exercise predicts automated IC performance prior to cGMP implementation, and the chemical equilibria framework predicts buffer conductivity with good agreement to actual conductivity measurements, as seen in **Figure 10** (on page 6). This further demonstrates that sound understanding of the chemical principles fundamental to automated IC performance is required for success. The critical parameters, and their proper controls required to ensure automated IC performance acceptable in a regulated cGMP environment, have been identified based on the experimental outcomes gathered in this study.

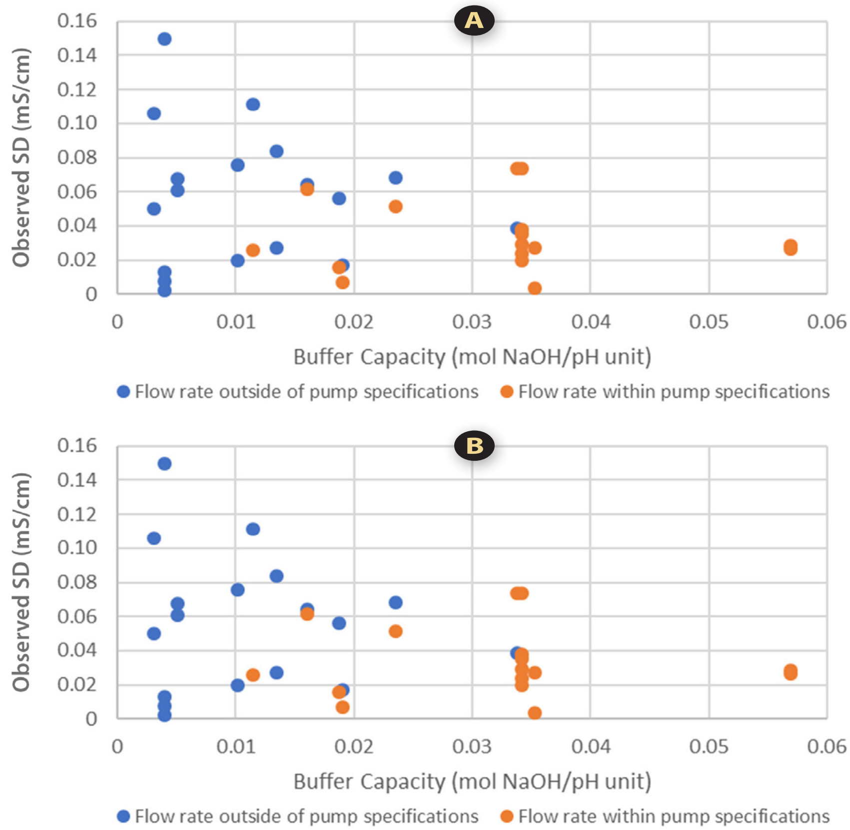

FIGURE 7. Conductivity variability vs. buffer capacity under: (A) pH-flow; and (B) flow feedback controls.

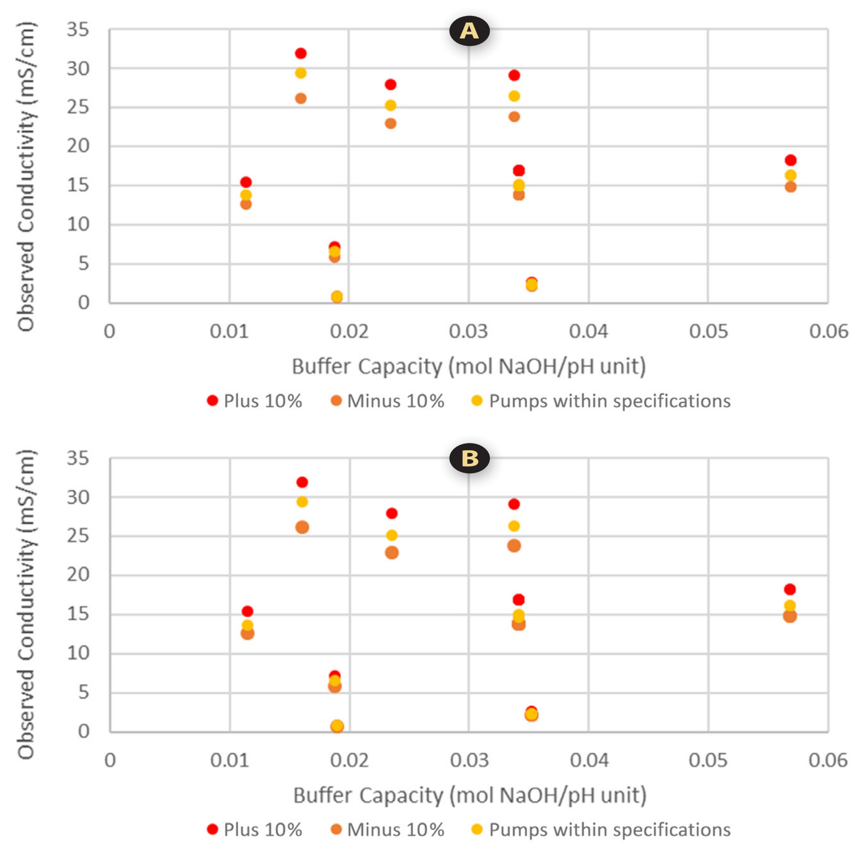

FIGURE 8. Maximum conductivity and ±10% limits vs. buffer capacity under: (A) pH-flow; and (B) flow feedback controls.

5. Discussions and Conclusions

In this paper, the ability to perform automated JIT at POU buffer management has been demonstrated using the IC technology described in Section 3. Critical parameters determining the robustness and reliability of this approach have been identified based on understanding of engineering design principles and chemical equilibria corrected for finite solution effect using the Debye-Hückel theory. Experiments were performed to evaluate the effects of buffer concentration, pH, salt concentration and flow rate on IC buffer mixing performance. Analysis of the experimental data identified: (1) buffer capacity; (2) operating flow rate with respect to pump design specifications; and (3) design of the feedback control scheme as the critical parameters impacting automated IC performance and robustness.

Findings from these experiments can be interpreted by accounting for sources of variability when mixing a buffer using the automated IC. Total system variability is accumulated over all variabilities originating from critical instrumentation such as pumps and meters for pH and conductivity. Such variabilities can be minimized by operating within the design specifications of each instrument which are inherent in an automated system. A buffer designed with high buffer capacity is least sensitive to the effects of normal system variability on target pH and is therefore best suited for preparation by an automated buffer mixing technology.

Under the specific scenarios investigated in this paper, automated IC likely reached the target pH with minimum iterations of feedback adjustment, as evidenced by the low pH variability whenever a high buffer capacity buffer was prepared using a pH-flow feedback control scheme. This "efficiency" also results in low conductivity variability because once the target pH is reached through minimum feedback iterations, the conductivity of the system, as defined by chemical equilibria, is also reached automatically. The effects of system specifications explicitly tested for pump flow rates applies equally to the pH and conductivity meters (not tested in this study), in that the target pH must fall within the linear range of the controlling pH probe throughout the run when a pH control loop is used. At the same time, operating within the conductivity meter specifications becomes less important, from a control perspective, because target buffer conductivity is reached automatically when the target pH is reached, but remains relevant from a data collection and process monitoring perspective. Since buffer capacity is a critical parameter to be controlled when automated IC is deployed, a valid model such as the one presented in this paper, shows good agreements between predicted and actual conductivity (**Appendix, Table 6**) and should be used to predict buffer capacity when designing a buffer to be formulated in an automated IC system.

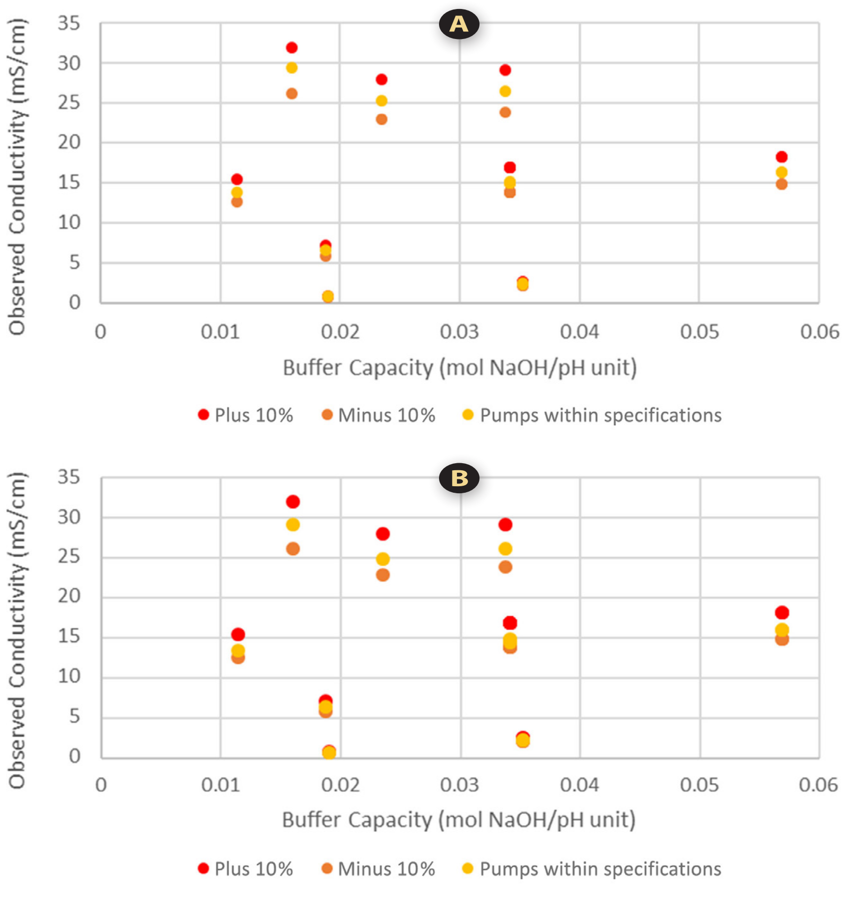

FIGURE 9. Minimum conductivity and ±10% limits vs. buffer capacity under: (A) pH-flow; and (B) flow feedback controls.

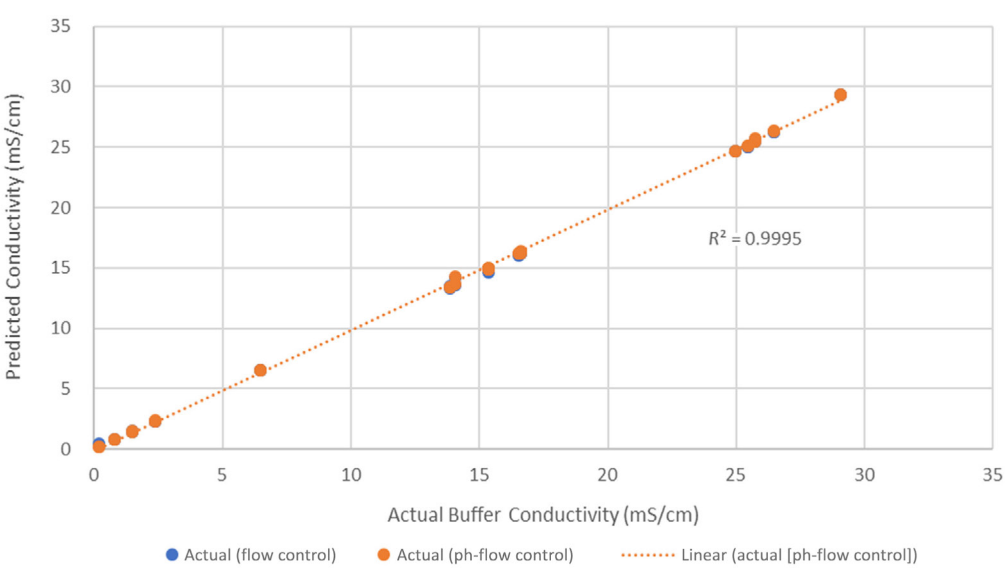

FIGURE 10. Calculated vs. actual conductivity measurements for solutions listed in **Table 6 (Appendix)** under flow and pH-flow feedback controls.

As discussed earlier, automated buffer mixing is deployed to streamline buffer preparation and storage operations. In addition to eliminating the need for buffer storage tanks, under the JIT at POU buffer formulation scenario, automated IC implemented with conjugate acid-base pairs can drastically reduce documentation and QC testing requirements. This is because only a handful of conjugate acid-base pair buffers need to be prepared to serve as universal starting materials for the entire range of buffers (*e.g.*, acetate) required for plant-wide downstream processing. Furthermore, the use of conjugate acid-base pairs guarantees fast convergence toward the target pH since each species at the point of mixing with WFI is close to its equilibrium concentrations. Under certain circumstances, outsourcing preparation of such conjugate acid-base pair solutions could be justified, allowing reassignment of personnel to execute higher-value activities that are more directly related to product production. An organization may choose to operate an IC-type technology in the in-line dilution (ID) mode to prepare a concentrated version of the final buffer, resulting in reduced buffer storage burdens. Such solutions could then be mixed with the appropriate amount of WFI at POU or into a hold vessel to be released through routine QC testing.

The critical parameters identified in this study governing effective automated IC implementation still apply for ID implementation since a solution prepared by ID is defined by the same chemical equilibria and engineering design principles. However, a plant operating in the ID mode will be preparing the same number of buffers, although at higher concentrations, and will not see the same benefits that JIT at POU provides.

Based on the theoretical and experimental data presented in this paper, parameters critical to automated IC system performance have been identified. When implemented properly, an automated buffer mixing technology such as IC has been demonstrated to be equivalent to manual buffer preparation in that it can consistently deliver buffers suitable for JIT at POU in biopharmaceutical manufacturing. The scientific and technical findings in this paper show that incorporating automated buffer mixing into licensed commercial processes is feasible when based on the following change management road map:

(1) Identify sources of variability embedded in the chosen automated buffer mixing technology.

(2) Properly construct experiments to identify critical parameters that impact system performance.

(3) Confirm system performance by revalidating the individual chromatography unit operation.

(4) Perform continued system performance verification through routine pH and conductivity monitoring.

(5) Mitigate normal pH and/or conductivity excursions with justifiable time averaging.

As described earlier, a review of the business drivers in biopharmaceutical downstream processing led the team to investigate and confirm the capabilities and validity of automated buffer mixing technologies. An understanding of chemical equilibria and engineering design principles formed the foundation of our automated buffer mixing deployment. An implementation plan based on well-designed experimental data has been proposed to guide the quality, compliance, and regulatory discussions and strategies required to streamline buffer handling. Commercial-scale downstream operations will be rewarded with lower productions costs in the high-stakes biopharmaceutical industry, which is under constant pressure to improve operating efficiency and deliver safe and affordable therapeutics for patients.

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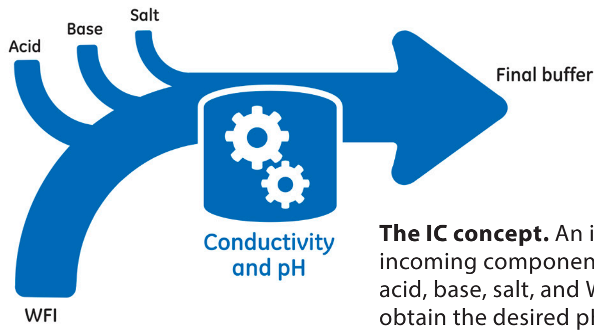
Ioana Erlandsson, GE Healthcare Bio-Sciences AB, presented portions of this article at the ISBioTech Waterside Conference held in Norfolk, Virginia USA, November 12–14, 2018. Presentation Title: "Deploying Automated Buffer Production for cGMP Use"

Appendix



The IC concept. An illustration of the incoming components to the IC system: acid, base, salt, and WFI are mixed to obtain the desired pH and conductivity.

TABLE 2. Maximum and minimum pH, and standard deviation under flow and pH-flow control presented in Figures 3-5. (Entries highlighted in green are experiments performed with pumps operating within design specifications.)

				pH-Flow Control					Flow Control				
pH	Buffer (M)	[NaCl] (M)	Buffer Capacity	SD	Max	Max – Target	Min	Min – Target	SD	Max	Max – Target	Min	Min – Target
3.7	0.02	0	0.00	0.05	3.74	0.04	3.59	-0.11	0.02	3.81	0.11	3.74	0.04
4.7	0.06	0.125	0.03	0.01	4.68	-0.02	4.66	-0.04	0.01	4.62	-0.08	4.6	-0.1
3.7	0.06	0.125	0.01	0.07	3.89	0.19	3.61	-0.09	0.03	3.83	0.13	3.73	0.03
5.7	0.06	0.125	0.01	0.08	5.82	0.12	5.52	-0.18	0.02	5.6	-0.1	5.52	-0.18
4.7	0.02	0.125	0.01	0.03	4.89	0.19	4.77	0.07	0.03	4.99	0.29	4.88	0.18
3.7	0.1	0	0.02	0.01	3.72	0.02	3.67	-0.03	0.01	3.69	-0.01	3.64	-0.06
4.7	0.06	0.125	0.03	0.01	4.72	0.02	4.68	-0.02	0.02	4.6	-0.1	4.55	-0.15
3.7	0.02	0.25	0.01	0.04	3.77	0.07	3.62	-0.08	0.03	3.85	0.15	3.7	0
5.7	0.1	0	0.02	0.02	5.71	0.01	5.65	-0.05	0.01	5.68	-0.02	5.63	-0.07
4.7	0.06	0.125	0.03	0.01	4.7	0	4.68	-0.02	0.01	4.66	-0.04	4.64	-0.06
5.7	0.02	0	0.00	0.27	6.02	0.32	5.26	-0.44	0.02	5.29	-0.41	5.2	-0.5
3.7	0.1	0.25	0.02	0.01	3.65	-0.05	3.58	-0.12	0.01	3.65	-0.05	3.58	-0.12
5.7	0.02	0.25	0.00	0.23	6.03	0.33	5.32	-0.38	0.07	5.49	-0.21	5.23	-0.47
4.7	0.06	0.25	0.03	0.01	4.7	0	4.68	-0.02	0.01	4.64	-0.06	4.6	-0.1
5.7	0.1	0.25	0.02	0.02	5.74	0.04	5.65	-0.05	0.02	5.67	-0.03	5.61	-0.09
4.7	0.1	0.125	0.06	0.00	4.7	0	4.68	-0.02	0.01	4.66	-0.04	4.64	-0.06
4.7	0.06	0	0.04	0.00	4.7	0	4.69	-0.01	0.01	4.69	-0.01	4.67	-0.03
3.7	0.02	0	0.00	0.04	3.62	-0.08	3.47	-0.23	0.02	4.18	0.48	4.11	0.41
4.7	0.06	0.125	0.03	0.00	4.71	0.01	4.69	-0.01	0.00	4.68	-0.02	4.66	-0.04
3.7	0.06	0.125	0.01	0.01	3.72	0.02	3.69	-0.01	0.01	3.66	-0.04	3.62	-0.08
5.7	0.06	0.125	0.01	0.01	5.74	0.04	5.69	-0.01	0.01	5.68	-0.02	5.65	-0.05
4.7	0.02	0.125	0.01	0.01	4.72	0.02	4.7	0	0.01	4.68	-0.02	4.66	-0.04
3.7	0.1	0	0.02	0.01	3.68	-0.02	3.66	-0.04	0.01	3.67	-0.03	3.65	-0.05
4.7	0.06	0.125	0.03	0.00	4.71	0.01	4.69	-0.01	0.01	4.7	0	4.67	-0.03
3.7	0.02	0.25	0.01	0.05	3.76	0.06	3.59	-0.11	0.01	3.65	-0.05	3.59	-0.11
5.7	0.1	0	0.02	0.01	5.69	-0.01	5.66	-0.04	0.01	5.7	0	5.67	-0.03
4.7	0.06	0.125	0.03	0.00	4.71	0.01	4.69	-0.01	0.00	4.66	-0.04	4.65	-0.05
5.7	0.02	0	0.00	0.01	5.68	-0.02	5.62	-0.08	0.02	5.71	0.01	5.65	-0.05
3.7	0.1	0.25	0.02	0.01	3.73	0.03	3.7	0	0.01	3.64	-0.06	3.61	-0.09
5.7	0.02	0.25	0.00	0.05	5.79	0.09	5.63	-0.07	0.02	5.68	-0.02	5.61	-0.09
4.7	0.06	0.25	0.03	0.01	4.7	0	4.69	-0.01	0.00	4.63	-0.07	4.62	-0.08
5.7	0.1	0.25	0.02	0.01	5.72	0.02	5.67	-0.03	0.01	5.65	-0.05	5.63	-0.07
4.7	0.1	0.125	0.06	0.00	4.7	0	4.69	-0.01	0.01	4.67	-0.03	4.65	-0.05
4.7	0.06	0	0.04	0.00	4.71	0.01	4.7	0	0.00	4.7	0	4.69	-0.01

TABLE 3. Conductivity (mSm^2mol^{-1}) maximum, minimum, and standard deviation under flow and pH-flow control presented in Figures 6-8. (Entries highlighted in green are experiments performed with pumps operating within design specifications.)

				Flow Control			pH-Flow Control				
pH	Buffer (M)	[NaCl] (M)	Buffer Capacity	SD	Max	Min	SD	Max	Min	Target + 10%	Target – 10%
3.7	0.02	0	0.00	0.01	0.26	0.24	0.01	0.24	0.21	0.22	0.18
4.7	0.06	0.125	0.03	0.03	14.94	14.81	0.04	15.03	14.88	16.89	13.82
3.7	0.06	0.125	0.01	0.05	13.59	13.38	0.08	13.59	13.28	15.23	12.46
5.7	0.06	0.125	0.01	0.05	16.33	16.12	0.08	16.56	16.25	18.28	14.96
4.7	0.02	0.125	0.01	0.11	14.47	14.03	0.11	14.49	14.06	15.48	12.67
3.7	0.1	0	0.02	0.02	0.82	0.75	0.02	0.84	0.78	0.86	0.71
4.7	0.06	0.125	0.03	0.06	14.75	14.51	0.07	15.12	14.79	16.89	13.82
3.7	0.02	0.25	0.01	0.05	24.74	24.53	0.06	24.86	24.58	27.45	22.46
5.7	0.1	0	0.02	0.05	6.66	6.44	0.06	6.68	6.45	7.14	5.84
4.7	0.06	0.125	0.03	0.03	14.91	14.78	0.04	15.04	14.88	16.89	13.82
5.7	0.02	0	0.00	0.07	1.61	1.35	0.15	1.89	1.32	1.64	1.34
3.7	0.1	0.25	0.02	0.08	25.22	24.86	0.07	25.21	24.95	27.98	22.89
5.7	0.02	0.25	0.00	0.15	25.93	25.34	0.11	25.93	25.49	28.31	23.16
4.7	0.06	0.25	0.03	0.06	26.36	26.12	0.04	26.44	26.26	29.11	23.82
5.7	0.1	0.25	0.02	0.07	29.49	29.18	0.06	29.57	29.28	31.95	26.14
4.7	0.1	0.125	0.06	0.04	16.17	16.01	0.03	16.29	16.17	18.20	14.89
4.7	0.06	0	0.04	0.03	2.39	2.27	0.03	2.43	2.31	2.61	2.14
3.7	0.02	0	0.00	0.02	0.47	0.41	0.00	0.22	0.21	0.22	0.18
4.7	0.06	0.125	0.03	0.03	14.92	14.81	0.03	14.97	14.83	16.89	13.82
3.7	0.06	0.125	0.01	0.02	13.38	13.30	0.03	13.45	13.31	15.23	12.46
5.7	0.06	0.125	0.01	0.03	16.31	16.16	0.02	16.31	16.22	18.28	14.96
4.7	0.02	0.125	0.01	0.02	13.63	13.52	0.03	13.76	13.65	15.48	12.67
3.7	0.1	0	0.02	0.01	0.80	0.78	0.01	0.81	0.79	0.86	0.71
4.7	0.06	0.125	0.03	0.03	15.00	14.86	0.02	14.99	14.89	16.89	13.82
3.7	0.02	0.25	0.01	0.06	24.78	24.51	0.07	24.82	24.51	27.45	22.46
5.7	0.1	0	0.02	0.02	6.59	6.49	0.02	6.54	6.47	7.14	5.84
4.7	0.06	0.125	0.03	0.03	14.92	14.81	0.02	15.00	14.91	16.89	13.82
5.7	0.02	0	0.00	0.01	1.48	1.42	0.01	1.47	1.42	1.64	1.34
3.7	0.1	0.25	0.02	0.07	25.10	24.84	0.05	25.28	24.99	27.98	22.89
5.7	0.02	0.25	0.00	0.05	25.56	25.36	0.05	25.58	25.38	28.31	23.16
4.7	0.06	0.25	0.03	0.06	26.38	26.12	0.07	26.51	26.20	29.11	23.82
5.7	0.1	0.25	0.02	0.05	29.42	29.20	0.06	29.46	29.19	31.95	26.14
4.7	0.1	0.125	0.06	0.03	16.14	16.01	0.03	16.29	16.15	18.20	14.89
4.7	0.06	0	0.04	0.01	2.34	2.30	0.00	2.36	2.34	2.61	2.14

Chemical Equilibria Revisited

Buffer conductivity can be predicted by summing up the conductivity of individual ionic species after solving for the equilibrium concentrations of weak electrolytes present in a solution. In this approach, the Kohlrausch equation[2] is used to calculate the molar conductivity of a subspecies of a weak electrolyte by applying the Kohlrausch coefficient, a fitted parameter obtained by fitting measured conductivities to the equation below:

$$\Lambda_i = \Lambda_{0i} - K_i * \sqrt{[c_i]} \qquad \text{(Eq. 1)}$$

wherein: Λ_i is the molar conductivity of the ionic species i

$\Lambda_0 i$ is the molar conductivity at infinite dilution of the ionic species i

$[c_i]$ is the concentration of the ionic species i

K_i is the Kohlrausch coefficient

While Λ_{0i} could be obtained from literature, both K_i and Λ_{0i} are usually obtained from data fitting of Eq. 1 at Λ_i and $[c_i]$.

The conductivity κ_i for each ionic species can then be calculated by the formula:

$$\kappa_i = [c_i] * \Lambda_i \qquad \text{(Eq. 2)}$$

Combining Eq. 1 with Eq. 2 gives:

$$\kappa_i = [c_i] * \{\Lambda_o - K * \sqrt{[c_i]}\} \qquad \text{(Eq. 3)}$$

Summation of individual conductivity for each ionic species *i* produces a predicted conductivity of the liquid solution. According to Eq. 1 above, the molar conductivity Λ_i of every species is approximately constant in the low concentration regime and decreases as a function of the square root of the concentration in a linear fashion. As the concentration increases, the conductivity κ_i increases asymptotically, instead of linearly, according to Eq. 3 due to a decrease in ion activity. This decrease can be accounted for by the Debye-Hückel theory[3] in Eq. 3 to obtain c_i. In this approach, the ionic strength of each species becomes a weighing parameter in calculating an average hydrated radius of the ionic species (an important parameter of the Debye-Hückel equation). Details of this approach are described below.

Buffer pH is governed by the Henderson-Hasselbach equation.[4] For a basic species (which may be a base B or a conjugate base A^-) in equilibrium with a corresponding acidic species (which may be a conjugate acid BH^+ or an acid HA, respectively), the solution pH is:

$$pH = pK_a + \log\left\{\frac{(Basic\ Species)}{(Acidic\ Species)}\right\} \qquad \text{(Eq. 4)}$$

In this equation, the parentheses denote the activity of each species rather than the concentration; the reason being that the ions involved tend to become shielded from the environment. The activity of each ion is related to the corresponding concentration through the activity coefficient φ:

$$(\text{species}) = \varphi\ [\text{species}] \qquad \text{(Eq. 5)}$$

In the ideal state of infinite dilution, φ becomes 1 and the activity of every ion will be equal to the corresponding concentration. However, in practice, the ionic strength is different from zero and the activity coefficients of the different species will be less than 1.

Inserting Eq. 5 into Eq. 4 above gives the pH in terms of the concentrations instead of activities:

$$pH = pK_a' + \log\left\{\frac{[Basic\ Species]}{[Acidic\ Species]}\right\} \qquad \text{(Eq. 6)}$$

where: $$pK_a' = pK_a + \log\varphi_a - \log\varphi_b \qquad \text{(Eq. 7)}$$

in which φ_a and φ_b are the activity coefficients for the acidic and basic species, respectively, and pK_a' is an apparent pK_a value which allows the use of the measurable concentrations of the different buffer species.

A model for these deviations is given in the so-called Debye-Hückel theory, known as:

$$\log\varphi = \frac{A * Z^2 I^{0.5}}{1 + 0.33 * 10^8 * \alpha * I^{0.5}} \qquad \text{(Eq. 8)}$$

wherein A is a temperature-dependent parameter with a value of ~ 0.51. Using data from literature[5, 6], the value of A can be calculated as:

$$A = (0.4918 + 0.0007) * T + 0.000004 * T^2 \qquad \text{(Eq. 9)}$$

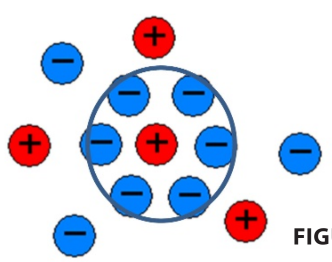


FIGURE 11.

where T is the temperature in degrees Celsius, Z is the charge of the ion and the quantity α, the ion size parameter, or the radii of the hydrated ions (in Å) described as the mean distance of approach of the ions due to finite concentration effect, as depicted in **Figure 11**. In turn, the value of pK_a' can be calculated by inserting Eq. 8 into Eq. 7:

$$pK_a' = pK_a + \frac{A*{Z_a}^2 I^{0.5}}{1+0.33*10^{8}*\alpha_a*I^{0.5}} - \frac{A*{Z_b}^2 I^{0.5}}{1+0.33*10^{8}*\alpha_b*I^{0.5}} \qquad \text{(Eq. 10)}$$

where the subscripts a and b specify the parameters corresponding to the acid and the base respectively, Z_a = charge of acidic species, Z_b = charge of basic species, α_a = ion size parameter of the acidic species, α_b = ion size parameter of the basic species, and $I = ½\,\Sigma(c_i\, Z_i^2)$.

As an approximation, the parameter α can be set to 3 Å for all buffer molecules[7], leading to the simplified formula:

$$pK_a' = pK_a + \frac{A*{Z_a}^2 I^{0.5}}{1+I^{0.5}} - \frac{A*{Z_b}^2 I^{0.5}}{1+I^{0.5}} \qquad \text{(Eq. 11)}$$

Eq. 11 above is the formula for ionic strength correction usually found in the literature. Using Eq. 6 in combination with the equations for (*i*) the conservation of mass, (*ii*) the conservation of charge, and (*iii*) the water dissociation equilibrium, the concentrations of the acidic and basic species of a monoprotic buffer at equilibrium can be calculated.

However, many buffers are polyprotic (i.e., their buffer molecules can accept and donate more than one proton corresponding to more than one pK_a value). The number of species in such a buffer system is always one more than the number of pK_a values. The calculation of the number of moles in each of the protonation species is equivalent to solving the equilibrium equations of each of the species with the neighboring species with one more and/or one less proton and with the concentration of hydrogen atoms reflected in the solution pH.

Assume, for example, a tritropic buffer such as citrate or phosphate with three pK_a values. Four protonation states or species will be defined ($s1$, $s2$, $s3$, and $s4$) by the three pK_a values. Three equations (corresponding to three pK_a values) may then be derived directly from Eq. 6 above:

$$xx[i] = 10^{\mathrm{pH}-pK_a[i]} \qquad \text{(Eq. 12)}$$

where each i corresponds to each $pK_a[i]$ value (i = 1, 2, 3), and $xx\,[i]$ are the ratios between the concentrations of the corresponding base and the corresponding acid (i.e., $xx\,[1] = [s_2/s_1]$, $xx\,[2] = [s_3/s_2]$, $xx\,[3] = [s_4/s_3]$.

In addition to these three equations, an equation arises because of the conservation of mass:

$$[s_1] + [s_2] + [s_3] + [s_4] = \text{buffer concentration} \qquad \text{(Eq. 13)}$$

and the conservation of charge:

$$[H^+] - [OH^-] + \Sigma\ \text{specific charge}\ (s_i) - \text{titrant\_charge} * [\text{titrant}] - \text{spec\_charge(start\_species)} * [\text{start\_species}] = 0 \qquad \text{(Eq. 14)}$$

where: start_species is the protonation state of the buffer prior to mixing

spec_charge(start species) is the charge of the protonation state of the buffer prior to mixing titrant

This protonation state is determined by the number of counter-ions per buffer molecule because of the electroneutrality requirement of the solution. The minus sign before $[OH^-]$ is due to the minus sign of the charge of the OH^- ions, whereas the minus sign in front of the two last terms is due to the charge of the counter-ions of the titrant and the start_species respectively.

Finally, there is the water dissociation equilibrium:

$$[OH^-][H^+] = 10^{-14} \qquad \text{(Eq. 15)}$$

Equations 12–15 above imply that there are six equations with six unknowns (the four $[s_i]$, $[OH^-]$, and $[H^+]$) in the case of three pK_a values. As such, the equilibrium concentrations of the respective acidic and basic species are completely defined. For a monoprotic buffer, the calculations are simplified: s_1 corresponds to the acidic species, s_2 to the basic species, and the concentrations of s_3 and s_4 are set to zero.

A more accurate determination of the ion size parameter α in the Debye-Hückel from Eq. 7 above is described in Bjorkestan *et. al*.[8] where α is determined as the weighted mean ion size of all species contributing to the ionic strength of the liquid mixture, using the ionic strength as weighting parameter. The ion size parameter α is calculated as:

$$\alpha = \Sigma \frac{I_i \alpha_i}{I} \qquad \text{(Eq. 16)}$$

where I_i is the ionic strength, α_i is the ion size parameter of species i, and I is the total ionic strength defined by:

$$I = \frac{1}{2} \Sigma c_i {Z_i}^2 \qquad \text{(Eq. 17)}$$

where c_i is the concentration and Z_i is the charge of ion present in the solution (in units of electronic charge), which gives:

$$I_i = \frac{1}{2} c_i {Z_i}^2 \qquad \text{(Eq. 18)}$$

In this study, buffer capacity is calculated by solving Eqs. 12–15 twice with NaOH as titrant at a given pH as follows:

pH_i = target pH

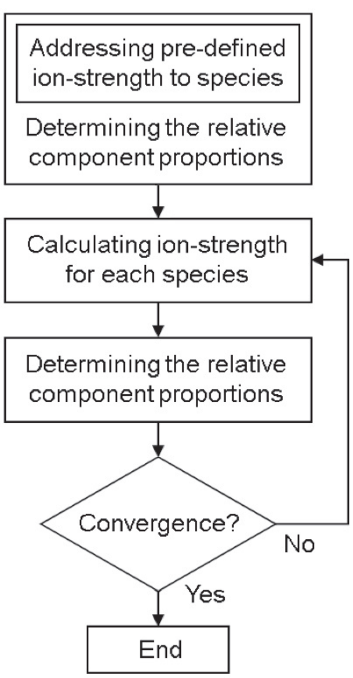
pH_{i+1} = target $pH + \delta\, pH$

where: $\delta\, pH = 0.005$

The buffer capacity is then calculated using the following formula:

$$\text{Buffer capacity} = \text{Abs}([\text{titrant}]_i - [\text{titrant}]_{i-1}) / \delta\, pH \qquad \text{(Eq. 19)}$$

Theoretical Prediction of Buffer Capacity, Target Buffer pH, and Conductivity



Since buffer capacity has been identified as one of the critical parameters in determining IC technology performance, the ability to accurately predict buffer capacity of a finite solution is important when implementing IC in a cGMP environment. Applying the equations developed in this section to each buffer specified in **Table 1**, individual buffer component concentrations required to maintain target pH and the conductivity, and subsequently the buffering capacity for each solution can be calculated. The validity of this approach and the model used is checked by comparing the DoE and calculated conductivity values. Using the acetate buffers listed in **Table 1**, for this study and assuming 22°C as the operating temperature, individual dissociation constants pK_a' were calculated through the algorithm below by solving Eqs. 11–15 for the equilibrium ionic concentrations c_i of acetate$^-$, Na^+, Cl^-, H^+ and OH^-.

The corresponding conductivity contributions were obtained using Eq. 3:

$$\kappa_i = [c_i] * \{\Lambda_o - K * \sqrt{[c_i]}\} \qquad \text{(Eq. 3)}$$

The total conductivity κ_{tot} for each buffer solution was obtained by summation of the conductivity of each ionic species:

$$\kappa_{tot} = \kappa_{Ac-} + \kappa_{Na+} + \kappa_{Cl-} + \kappa_{H+} + \kappa_{OH-}$$

Λ_0 (mSm^2mol^{-1}) in Eq. 3 for the respective species were obtained from[9] and are listed in **Table 4** on the following page.

TABLE 4. Infinite dilution molar conductivity coefficient Λ_0 for acetate-, Na^+, H^+, OH^-, and Cl^-.

Species	Λ_0 (mSm^2mol^{-1})
$Acetate^-$	40.9
Na^+	50.1
H^+	349.6
OH^-	199.1
Cl^-	76.3

TABLE 5. Kohlrausch coefficient for acetate-, Na^+, H^+, OH^-, and Cl^-.

	$Acetate^-$	Na^+	H^+	OH^-	Cl^-
Λ_0	40.9	50	350	199	76
K	17	49	0	0	6

Using Eq. 3 with the concentration values c_i and the Λ_0 in **Table 4**, a fitting procedure was used to obtain the respective values for the Kohlrausch coefficient K that gave the best fit with the measured conductivity values collected from the DoE experiments. The resulting K and Λ_0 values for $acetate^-$, Na^+, Cl^-, H^+, and OH^- are provided in **Table 5.**

As a result, contributions from different ions and the predicted total conductivity of the buffer solution combined with the measured conductivity values from **Table 3** are illustrated in **Table 6** within the appropriate buffer capacity regime. The measured conductivities under both flow feedback and pH-flow feedback controls are plotted against the predicted conductivity, as shown in **Figure 10**. The goodness of fit seen in **Figure 10** demonstrates that the procedure outlined in this section can be used to predict buffer conductivity accurately, and the model can be used to guide buffer design to ensure successful IC skid cGMP operations within the appropriate buffer capacity regime.

TABLE 6. Predicted and actual conductivity values (mSm^2mol^{-1}). (Entries highlighted in green are experiments performed with pumps operating within design specifications.)

pH	Buffer (M)	[NaCl] (M)	Buffer Capacity	Predicted	Actual (Flow Control)	Actual (pH-Flow Control)
3.7	0.02	0	0.00	0.20	0.25	0.22
4.7	0.06	0.125	0.03	15.35	14.87	14.96
3.7	0.06	0.125	0.01	13.85	13.46	13.42
5.7	0.06	0.125	0.01	16.62	16.21	16.39
4.7	0.02	0.125	0.01	14.07	14.20	14.28
3.7	0.1	0	0.02	0.79	0.79	0.81
4.7	0.06	0.125	0.03	15.35	14.64	14.94
3.7	0.02	0.25	0.01	24.96	24.65	24.70
5.7	0.1	0	0.02	6.49	6.57	6.57
4.7	0.06	0.125	0.03	15.35	14.85	14.97
5.7	0.02	0	0.00	1.49	1.48	1.48
3.7	0.1	0.25	0.02	25.43	25.02	25.09
5.7	0.02	0.25	0.00	25.73	25.60	25.72
4.7	0.06	0.25	0.03	26.47	26.22	26.32
5.7	0.1	0.25	0.02	29.05	29.34	29.38
4.7	0.1	0.125	0.06	16.54	16.09	16.24
4.7	0.06	0	0.04	2.38	2.33	2.38
3.7	0.02	0	0.00	0.20	0.45	0.21
4.7	0.06	0.125	0.03	15.35	14.85	14.91
3.7	0.06	0.125	0.01	13.85	13.34	13.37
5.7	0.06	0.125	0.01	16.62	16.23	16.26
4.7	0.02	0.125	0.01	14.07	13.58	13.70
3.7	0.1	0	0.02	0.79	0.79	0.80
4.7	0.06	0.125	0.03	15.35	14.93	14.94
3.7	0.02	0.25	0.01	24.96	24.64	24.66
5.7	0.1	0	0.02	6.49	6.54	6.51
4.7	0.06	0.125	0.03	15.35	14.87	14.96
5.7	0.02	0	0.00	1.49	1.45	1.45
3.7	0.1	0.25	0.02	25.43	25.00	25.11
5.7	0.02	0.25	0.00	25.73	25.48	25.49
4.7	0.06	0.25	0.03	26.47	26.23	26.34
5.7	0.1	0.25	0.02	29.05	29.29	29.32
4.7	0.1	0.125	0.06	16.54	16.07	16.20
4.7	0.06	0	0.04	2.38	2.32	2.35